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Too much light blinds: The transparency-resistance paradox in algorithmic management

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ABSTRACT

Gig platforms increasingly harness AI algorithms to manage workers, offering notable efficiency and scalability benefits. However, the rise of worker resistance, such as manipulating algorithms with fake data, poses challenges to these benefits. These algorithms are often perceived as ''black boxes'', leading to issues around transparency. This research thus explores the impact of algorithmic transparency on worker resistance. Using a longitudinal design, we uncovered a paradox: Initially, greater transparency correlates with enhanced fairness perception and reduced resistance. However, beyond a certain threshold, further transparency starts to backfire, leading to decreased fairness perception and amplified resistance. This paradox challenges the prevailing notion that more transparency always leads to positive outcomes. Moreover, we examined the role of human managers, showing that their empathetic support and caring can mitigate worker resistance when transparency fails to foster fairness. This highlights the power of human touch in the algorithm-driven workplace. Overall, these insights suggest a hybrid management model, wherein the cold efficiency of algorithmic managers is complemented by the warm empathy of human managers, offering a blueprint for more productive and harmonious human-machine interactions in the gig economy.

1. Introduction

In the gig economy, the role of AI algorithms in managing work has become increasingly pivotal. Platforms like Uber and Deliveroo employ algorithms to execute key managerial functions, including task allocation, process monitoring, and performance evaluation (Möhlmann et al., 2021). This algorithm-driven management enhances operational efficiency and scalability for platforms (Tomprou & Lee, 2022). However, it also creates a challenging work environment for workers on them. Particularly, workers often encounter algorithmic decisions that provide scant insight into decision-making processes (Langer & König, 2023). For instance, a ride-hailing driver might unexpectedly see a decrease in high-paying ride requests without a clear explanation. Similarly, a food-delivery rider may notice their assignments tend toward longer, less profitable routes, without any apparent logic (Heiland, 2023). These instances illustrate the everyday realities faced by gig workers, prompting us to consider how they respond to the transparency of algorithmic systems.

Notably, the rise of worker resistance to algorithmic systems is a growing concern for platform managers. This resistance manifests in various behaviors intended to counteract, mitigate, or otherwise resist the effects of algorithms, such as evading monitoring or providing false data (Cameron & Rahman, 2022). These behaviors can undermine the effectiveness and efficiency of algorithmic management. For instance, their evasion of monitoring and provision of false data can compromise the scope and accuracy of the data on which algorithms are based (Benlian et al., 2022). More importantly, worker resistance has negative implications for platform reputation and customer trust. For example, to resist algorithmic scoring, workers might negotiate privately with customers to alter negative reviews, potentially distorting the platform's reputation system and misleading other customers (Bellesia et al., 2023). Hence, it is imperative for platforms to effectively mitigate worker resistance to fully leverage the advantages of algorithmic management.

The challenge lies in understanding the triggers of these resistance behaviors, particularly in environments where algorithmic decision-making processes are obscured by their inherent complexity (Möhlmann et al., 2023). On the surface, acts of resistance may appear simply as expressions of distress (Pregenzer et al., 2021a). However, it is crucial to empirically investigate whether these behaviors also suggest a

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need among workers for a clearer understanding of the algorithmic systems that govern their work. In essence, this phenomenon reflects a tension between the algorithmic systems designed to optimize efficiency and the human elements of the workforce they govern. This tension motivates us to ask: How does algorithmic transparency impact workers' resistance behaviors on gig platforms?

Previous research has explored how workers perceive and react to the transparency of algorithmic systems, but yielded mixed findings. Some studies showed that transparency can facilitate workers make sense of algorithms, enhance their understanding and satisfaction with algorithmic decisions, decrease turnover intentions, and enable them to better navigate their roles within gig platforms (Bujold et al., 2022; Chen et al., 2023; Möhlmann et al., 2023; Ochmann et al., 2024; Rahman, 2021). Conversely, other studies indicated that increased transparency can lead to unintended negative outcomes. Excessive transparency may reduce advice utilization when algorithms are perceived as too simple, lead to information overload or misinterpretation, harm user trust in AI systems, and even cause individuals to modify their behavior based on inaccurate data (Ananny & Crawford, 2018; de Fine Licht & de Fine Licht, 2020; Schmidt et al., 2020; Lehmann et al., 2022; Bauer & Gill, 2024). Prior works have also explored how workers navigate and respond to the ''black box'' nature of these systems through various coping mechanisms, such as folk theories, reliance on pods, and information networks for best practices (Wood et al., 2018; O'Meara, 2019; Siles et al., 2020). These studies provide valuable insights into the diverse impacts of algorithmic transparency and advance our understanding of worker adaptations to algorithmic environments.

However, the existing literature primarily focuses on linear effects, either positive or negative, without delving into the possibility that transparency's impact on worker behavior could follow a nonlinear pattern. This study aims to fill this gap by exploring whether there is a curvilinear relationship between algorithmic transparency and worker resistance, identifying the points at which increased transparency no longer yields positive outcomes and may begin to yield diminishing or even adverse returns. Through examining this nuanced dynamic, our research intends to shed light on more sophisticated mechanisms of algorithmic management, potentially leading to improved practices that better balance the effectiveness of algorithmic systems and the well-being of workers governed by them.

This research makes two main contributions to the ongoing discourse on algorithmic management and worker behavior in the gig economy. First, contrary to widespread calls for greater transparency in algorithmic systems (Bitzer et al., 2023; Dekker et al., 2022; Sturm et al., 2023), our research unveils a paradox: although some transparency helps reduce worker resistance, beyond a certain threshold, further transparency can backfire, leading to intensified resistance. This challenges the prevailing notion that more transparency corresponds to positive outcomes (Wanner et al., 2022; Aysolmaz et al., 2023; Suen & Hung, 2023), suggesting that balanced transparency is essential for minimizing worker resistance. Second, this research is among the first to explore how human interventions shape workers' reactions to algorithmic systems. The role of manager caring in mitigating resistance underlines the irreplaceable value of human managers in providing the empathetic support that "algorithmic managers" lack. Thus, this work offers new insights into the synergy between humans and algorithms by advocating for a hybrid "human-algorithm co-management" model wherein human empathy complements algorithmic efficiency.

2. Theoretical background and hypotheses

2.1. Algorithmic management in the gig economy

The gig economy, characterized by its flexible, on-demand work structure, has become a significant part of the modern labor market. In this rapidly growing sector, platforms such as Uber and Deliveroo have redefined traditional employment models, leveraging digital platforms to connect freelance workers with short-term contracts. Central to the operation of these platforms is algorithmic management, a system where AI algorithms play a critical role in managing work. In this landscape, algorithmic systems are not just tools but act as virtual managers. They perform essential managerial functions traditionally done by humans, such as task allocation, process monitoring, and performance evaluation (Möhlmann et al., 2021). For instance, Uber algorithms determine the allocation of rides to drivers, monitor the process in real-time, and evaluate driver performance based on a variety of metrics like speed, efficiency, and customer feedback. Similarly, in food delivery platforms like Deliveroo, algorithms manage the logistics of order assignment, track the delivery process, and assess performance indicators such as delivery times and customer ratings (Parent-Rocheleau & Parker, 2022).

The benefits of algorithmic management are manifold, particularly in terms of efficiency and scalability. Algorithms can process vast amounts of data at an unprecedented speed, allowing for real-time decision-making that significantly enhances operational efficiency. For instance, in ride-hailing platforms, algorithms can quickly match drivers with passengers based on location, route, and traffic conditions, optimizing wait times and route efficiency (Benlian et al., 2022). This efficiency is crucial in the fast-paced gig economy, where customer satisfaction often hinges on the speed and reliability of the service. Furthermore, the scalability offered by algorithmic management is another standout advantage. It enables these platforms to rapidly expand their services to new markets and handle a growing number of transactions without the need for a proportional increase in managerial personnel (Tomprou & Lee, 2022). This scalability is evident in the ease with which these platforms can adapt to different regional demands and manage fluctuating market conditions, such as peak hours or special events. These benefits enable platforms to manage a vast network of gig workers across various regions, catering to a growing customer base with relative ease.

However, this algorithm-driven approach also brings significant challenges for workers, with transparency being a primary concern. While algorithms excel in efficiency, they often operate as ''black boxes'', with their decision-making processes being opaque to the workers. Workers often find themselves at the mercy of algorithms that dictate crucial aspects of their work, without clear insight into the logic and criteria behind these decisions (Langer & König, 2023). For example, a food-delivery rider might receive requests for short, less profitable orders despite being in a high-demand area, unable to discern the algorithm's criteria for order assignments. Similarly, a ride-hailing driver might experience sudden changes in surge pricing that are not clearly aligned with their understanding of peak hours. This opacity can result in the driver being unable to plan their work schedule to maximize earnings, leading to discontent and even a sense of being unfairly treated by the algorithms (Jabagi et al., 2024). These examples illustrate how the uncertainty inherent in algorithmic systems can affect gig workers' day-to-day work life and overall well-being. The challenge of transparency is thus a crucial issue that needs to be addressed to ensure not only efficiency but also a better human-machine experience.

Previous research has explored how workers perceive and react to the transparency of algorithmic management, but demonstrated mixed findings regarding the impacts of algorithmic transparency on worker experiences and behaviors. Some studies suggested positive outcomes associated with more transparency. For instance, Rader et al. (2018) shown that explanations within algorithmic process can significantly increase user understanding and satisfaction, while Bujold et al. (2022) reported that the transparency of algorithms systems in surveillance and performance management can reduce driver's turnover intention. Similarly, Chen et al. (2023) found that the transparency in algorithmic performance appraisal helps gig workers easily craft their platform work. Conversely, other studies point to the complexities and potential drawbacks when algorithmic systems becomes too transparent. Ananny and Crawford (2018) argued that transparency might not always build trust and can instead invoke neoliberal models of agency, potentially leading to negative outcomes. de Fine Licht and de Fine Licht (2020) further argued that while transparency is generally pursued to increase perceived legitimacy, it can paradoxically lead to decreased public trust in AI decision-making because too much transparency may not always be comprehensible to the general public. More recently, Bauer and Gill (2024) found that disclosing algorithmic scores to individuals can lead to self-fulfilling prophecies where people alter their behavior to align with the disclosed but fundamentally erroneous scores.

While these studies have laid a solid foundation in understanding the impact of algorithmic transparency, there remains a gap in exploring the potentially nonlinear effects of transparency on worker resistance. The existing literature often focuses on linear interpretations—either positive or negative—without fully addressing how transparency might have nonlinear effects on worker behaviors. Our research aims to fill this gap by empirically testing whether a curvilinear relationship exists between transparency and resistance. This study seeks to uncover the thresholds at which transparency begins to have diminishing returns or even negative effects on worker perceptions and actions. By focusing on this nonlinear dynamic, we hope to provide new insights that could inform the development of more balanced algorithmic management practices, enhancing both the effectiveness of these systems and the well-being of workers governed by them.

2.2. Worker resistance against algorithmic management

Worker resistance refers to a range of actions undertaken by employees to oppose or counteract policies, practices, or conditions imposed by their employers or management systems (Mumby et al., 2017). In our context, we define it as actions that workers undertake with the intent to subvert or undermine the algorithmic systems governing them. In the traditional management setting, these behaviors can range from subtle acts of non-compliance, such as slowing down work or bending rules, to more overt acts of defiance, such as strikes or sabotage. Previous research suggests that resistance behaviors can be more covert in the context of algorithmic management (Jiang, 2023). Gig workers are often classified as independent contractors rather than employees, lacking many of the formal channels of resistance available to traditional employees, such as unions or collective bargaining (McDaid et al., 2023). As a result, they may resort to new forms of resistance. For instance, workers might provide false data or exploit algorithmic loopholes to manipulate the systems in their favor, effectively resisting algorithmic surveillance (Wu et al., 2023). Similarly, workers might privately negotiate with customers to alter or remove negative reviews, thereby maintaining their high reputations on the platform (Bellesia et al., 2023). These acts of resistance are not mere expressions of discontent but are indicative of deeper issues related to the algorithmic governance of work.

Importantly, such resistance has become a growing concern for platform operators, negatively impacting the operational efficiency of gig platforms and potentially harming their reputation. From a practical standpoint, resistance can disrupt the algorithm's intended task allocation and efficiency, leading to suboptimal resource distribution and decreased productivity (Wiener et al., 2023). For example, evading algorithmic monitoring and providing fake data can compromise the scope and accuracy of the data on which algorithms are based. Moreover, worker resistance can have broader implications for platform reputation and customer trust. When workers negotiate with customers to alter or remove negative reviews, it can distort a platform's reputation system and potentially mislead other customers. This could lead to a loss of trust in the platform's ability to provide reliable and high-quality services, which is crucial for customer retention and loyalty. These negative implications highlight the need to understand and address the underlying causes of worker resistance, thereby sustaining the health and growth of the gig economy.

Recent studies have highlighted factors such as unpredictability in earnings, perceived biases in rating systems, and the impersonal nature of algorithmic feedback as significant contributors to workers' dissatisfaction and resistance (Vasudevan & Chan, 2022; Cini, 2023; McDaid et al., 2023). Prior works also explored how workers navigate and respond to the ''black box'' nature of these systems through various coping mechanisms, such as folk theories (Siles et al., 2020), reliance on pods (O'Meara, 2019), and information networks for best practices (Wood et al., 2018). These studies provide valuable insights into the diverse impacts of algorithmic management on worker experiences and advance our understanding of worker adaptation to algorithmic environments. Our research seeks to complement these insights by delving deeper into how the overarching framework of algorithmic transparency itself may influence these reactions. Specifically, given the mixed positive and negative findings in prior research as discussion above, we aims to empirically test the potential nonlinear effects of transparency and uncover at what points increasing transparency might begin to have diminishing returns or even counterproductive effects on worker resistance behaviors.

2.3. Algorithmic transparency and fairness

Algorithmic transparency refers to the clarity and accessibility of the processes and criteria used by algorithms in decision-making (Shin & Park, 2019). In the gig economy, this transparency involves revealing the mechanisms through which algorithms allocate tasks, evaluate performance, and determine compensation. Workers' reactions to algorithmic decisions are largely grounded in their assessment of fairness – whether these decisions are made in an unbiased and equitable manner (Lavanchy et al., 2023; Wang et al., 2023). Therefore, algorithmic fairness should be considered in exploring the impact of algorithmic transparency on worker behavior. Intuitively, increased transparency would lead to heightened perceptions of fairness, as workers have clearer insights into decision-making process and criterion. However, this assumption does not always hold. The inherent complexity of AI algorithms can make them difficult to understand, even when their inner workings are fully disclosed (de Fine Licht & de Fine Licht, 2020). Workers might see how decisions are being made; however, this visibility does not necessarily equip them to effectively understand and assess these processes (Ananny & Crawford, 2018). In other words, increased transparency does indeed provide workers with more information about the algorithmic processes. Yet, this does not automatically translate into an increased sense of fairness. Therefore, while transparency and fairness are closely related concepts, the relationship between them may be more intricate than it appears.

Social Exchange Theory (SET) provides a useful lens to advance our understanding of the dynamics between algorithmic transparency and fairness. The application of SET is premised on the Computers Are Social Actors (CASA) paradigm (Nass et al., 1994; Nass & Moon, 2000). CASA suggests that people often engage with computers—and by extension, algorithms—as if they were social beings, applying social rules and expectations mindlessly, even though they are aware that these entities do not possess human motivations or emotions. This mindless anthropomorphism means that workers can perceive algorithmic systems through the lens of social exchange, evaluating fairness and reciprocity as they would in human interactions. The CASA paradigm has been validated in numerous studies (Kang & Gratch, 2014; Karr-Wisniewski & Prietula, 2010; Thuillard et al., 2022; Wang, 2017), supporting the application of SET to understand how workers interact with algorithmic systems, despite these systems' lack of inherent sociality or personality.

SET posits that the essence of social interactions is the process of social exchange. Individuals incur certain costs during these interactions, and in exchange, they expect corresponding benefits (Cropanzano & Mitchell, 2005). The costs involved can be tangible, such as time and physical labor, or intangible, like emotional effort and cognitive processing. Similarly, benefits can also be tangible, including financial rewards and material goods, or intangible, such as emotional satisfaction and social status (Cropanzano, Anthony, et al., 2017a,b).

People seek fair treatment in their social interactions, and the fairness perception is based on their assessment and comparison of costs and benefits. If individuals feel that their input (costs) has been adequately or more than adequately compensated with returns (benefits), they perceive the situation as fair. Conversely, if the input seems to exceed the returns, feelings of unfairness arise (Lavelle et al., 2009).

In gig platforms, algorithms take on the role of a virtual manager, while workers invest time and effort (costs) to complete tasks assigned by these algorithms, in exchange for monetary compensation and more job opportunities (benefits). In this regard, the interaction between workers and algorithms can be viewed as a social exchange process. Importantly, the workers' costs extend beyond mere time and labor. They also include the psychological stress of adapting to algorithm-driven decisions and the cognitive effort needed to understand intricate algorithmic processes (You et al., 2022). On the benefit side, beyond financial gains, workers can achieve a sense of satisfaction by effectively navigating and utilizing these algorithmic systems, coupled with the fulfillment that comes from enhancing their professional skills (Lee, 2018). Ultimately, workers' perception of fairness in the gig economy is shaped by their assessment of the trade-off between the costs they bear and the benefits they accrue.

In algorithmic management, workers often encounter ''black box'' decision-making processes. In this case, the transparency of these algorithms plays a pivotal role in the social exchange between workers and algorithms, influencing their perception of fairness in the balance between work input and return. When transparency is low, employees have limited insight into how decisions are made and resources are allocated, leading to significant uncertainty (Heiland, 2023). For instance, a delivery worker may not understand why they get fewer orders than others. As transparency increases, such as when employees are shown how tasks are assigned based on customer ratings and past performance, this uncertainty can be reduced (Liu, 2021). They start to see a clearer picture of how decisions are made, feeling that their efforts are more justly evaluated and rewarded. This enhanced understanding can lower psychological costs, like skepticism and distrust towards algorithmic decisions, while increasing perceived benefits, such as a sense of control over their work and job satisfaction, thereby increasing their sense of fairness (Bujold et al., 2022).

When transparency reaches a critical point, workers' sense of fairness may peak. This critical point refers to the level of transparency that is just sufficient for workers to fully understand their work conditions and the logic of algorithmic decisions, without causing information overload. At this point, workers can clearly see how their efforts affect work outcomes, with their cognitive costs (e.g., the effort to understand algorithmic logic) and emotional costs (e.g., dealing with uncertainty and insecurity) being at the lowest (Parent-Rocheleau & Parker, 2022). Concurrently, their perceived benefits, including mastery over work processes and job satisfaction, can be at their highest (Kinowska & Sienkiewicz, 2023). Thus, at this critical point, workers may feel that there is an optimal balance between their efforts and the returns they receive, leading to the highest level of fairness perception.

However, when transparency exceeds this critical point, further increasing it may reduce employees' sense of fairness. This is mainly because excessive transparency not only increases the cognitive burden on employees but also reduces their perceived benefits (You et al., 2022). For example, if the algorithms provide overly complex or technical data, it may exceed the workers' understanding and application scope. In such cases, worker face significant increases in psychological costs, such as the difficulty of processing excessive information and the resulting emotional stress (Vimalkumar et al., 2021). Meanwhile, as the information becomes difficult to understand and apply, workers might feel that their control over the work environment decreases, reducing their understanding of work processes and their influence on outcomes. Therefore, although increased transparency provides more information, its excessive complexity reduces employees' sense of control and job satisfaction, thereby reducing their perceived benefits (Möhlmann et al., 2023). In this situation, the enhancement of transparency not only increases costs but also reduces benefits, leading to a decrease in workers' sense of fairness. Transparency may no longer serve as a contributing factor but rather as a burden that exacerbates feelings of unfairness. Therefore, we hypothesize that.

H1. The relationship between algorithm transparency and fairness resembles an inverted U-shaped curve. Initially, heightened transparency may enhance perceived fairness; however, beyond a certain threshold, additional transparency may reduce perceived fairness.

2.4. Algorithmic fairness and worker resistance

Algorithmic fairness in the gig economy is a concept that captures how equitable and impartial workers perceive the algorithms governing task allocation, performance evaluation, and compensation (Lee, 2018). This encompasses not only the fairness of the outcomes but also the fairness of the methods used to arrive at these outcomes. A worker's sense of fairness, or the lack thereof, can profoundly influence their work motivation, job satisfaction, and willingness to comply with the algorithms (Langer & Landers, 2021). For instance, consider a ride-sharing platform where drivers perceive the allocation of rides and earnings calculation as fair. This perception likely stems from their understanding that the algorithm allocates rides based on equitable factors like proximity and driver availability, and calculates earnings transparently. In such a scenario, drivers are more inclined to trust the algorithms, feel valued, and are less likely to resist through means such as logging off during peak hours or ignoring ride requests (Höddinghaus et al., 2021).

Prior studies suggested that fairness perception is critical in shaping workers' attitudes and behaviors, including their propensity to resist or accept the system (Hellwig et al., 2023; Shanahan & Smith, 2021; Starke et al., 2022). When workers believe that the algorithmic systems governing their tasks are fair, they tend to exhibit positive behaviors such as higher engagement and compliance. This positive reaction is rooted in the belief that the system is treating them justly and equitably, leading to satisfaction and loyalty towards the platform (Shanahan & Smith, 2021). For instance, when a ride-hailing app equitably distributes high-demand rides among drivers, or when a delivery service fairly compensates for distance covered, workers are more likely to feel fairly treated and show less resistance to the platform's operational model. This sense of fairness can extend beyond financial compensation to include aspects like transparent communication, equitable opportunity for high-earning tasks, and fair resolution of disputes (Hellwig et al., 2023). Such an environment not only boosts morale but also encourages a more cooperative and productive relationship between workers and platforms.

On the contrary, the perceptions of low algorithmic fairness may lead to increased resistance among workers. This resistance manifests in a spectrum of behaviors, from subtle to overt. Subtle forms of resistance include decreased enthusiasm and effort in tasks (Vasudevan & Chan, 2022). For example, a delivery rider might opt for longer routes or take more breaks, subtly undermining the platform's service efficiency. In more overt forms, workers might protest platform policies or voice their grievances on social media and public forums (Wood et al., 2023). Such actions often arise from various aspects of perceived unfairness, such as inequitable task distribution, biases in performance evaluations, or opaque compensation calculations. Workers may feel that their efforts are not being justly rewarded, or that the algorithmic system is biased against certain groups of workers (Kordzadeh & Ghasemaghaei, 2022). These perceptions of injustice can lead to a breakdown in the relationship between workers and the platform, propelling them to resist as a means of expressing their discontent and pushing for change. Thereby, we speculate that.

H2. Algorithmic fairness is negatively associated with worker resistance, such that greater perceived fairness in algorithmic decisions correlates with lesser resistance among workers.

2.5. The buffering role of manager caring

In the rapidly evolving landscape of the gig economy, the role of human managers is undergoing significant changes. While "algorithmic managers" are increasingly taking over the task of managing workers, the significance of human managers still cannot be neglected (Jarrahi et al., 2023). Indeed, they serve as a bridge between the platform and the workers, providing humanistic caring that "algorithmic managers" cannot offer. For instance, Meituan, China's largest food delivery platform, integrates human managers into its operational framework by setting a station master at each delivery station, acting as a supervisor to directly manage delivery riders within the station (Wang et al., 2021). These supervisors provide more than just oversight; they can offer a level of caring and understanding that algorithms inherently lack (Wu et al., 2023). This caring involves attending to the emotional and personal needs of workers, fostering a supportive work environment.

Manager caring refers to the display of empathy, support, and understanding by managerial staff towards their employees (Micevski et al., 2017). This concept involves more than just addressing work-related issues; it extends to acknowledging the personal challenges and individual needs of workers. Manager caring is characterized by actions that demonstrate genuine concern for the well-being of employees, such as offering individualized support, actively listening to and addressing their concerns, and recognizing their efforts and contributions (Saks, 2022). This approach can be particularly effective in helping workers adapt to and accept the decisions made by algorithms, especially when these decisions impact their work or sense of fairness, as it adds a human touch to the impersonal nature of algorithmic decision-making (Wu et al., 2023). Therefore, from the standpoint of "Human-Algorithm Co-management", we argue that manager caring should be considered when examining workers' reactions to algorithmic management in gig platforms.

Besides the practical significance stated above, the inclusion of manager caring in our research is also motivated by theoretical considerations. Affective events theory posits that specific events at work (affective events) trigger emotional responses in employees, which in turn influence their work attitudes and behaviors (Mignonac & Herrbach, 2004). For instance, an employee who feels joy and pride from receiving recognition might exhibit increased job engagement. On the contrary, experiencing anxiety due to an excessive workload might lead to reduced job performance or resistance to organizational policies. These affective events can be either positive (such as being assigned a desirable project or receiving public acknowledgment from management) or negative (such as encountering technical glitches in work systems or facing abrupt schedule changes), shaping different emotional states of employees, thereby impacting their job satisfaction, engagement, and performance (Carlson et al., 2011).

In the context of algorithmic management, the caring from human managers can be viewed as a positive affective event due to its inherent nature of addressing and nurturing the emotional and personal needs of employees (Saks, 2022). When managers show genuine concern, empathy, and support, these actions evoke positive emotional responses in employees, because such caring behaviors make them feel valued, respected, and supported (Cropanzano, Anthony, et al., 2017a,b). These emotions can significantly influence how employees perceive their work environment and their responses to it. Therefore, manager caring as a positive event may moderate employees' reactions to negative events or perceptions at work, such as those stemming from algorithmic decisions perceived as unfair. We now turn to a more detailed discussion on the buffering role of manager caring.

As discussed in the previous section, when workers perceive a lack of fairness in algorithmic management, they may experience frustration or resentment, leading to resistance behaviors (Möhlmann et al., 2021). However, the presence of caring managers can alleviate these negative feelings by providing emotional support, offering a listening ear, and engaging in resolving workers' concerns. For example, a manager might patiently explain the rationale behind the algorithmic system, attentively listen to the worker's concerns, or even advocate for the worker's grievances with the platform. These caring behaviors can help workers feel understood and valued, thereby reducing their inclination to resist the algorithms (Eisenberger et al., 2020). As a result, even in situations where unfairness is perceived, workers may feel less compelled to resist the algorithms because manager caring provides an alternative channel for expressing and resolving grievances, thereby diminishing the necessity to engage in resistance as a form of expressing discontent.

Conversely, in the absence of manager caring, the negative impact of perceived fairness on worker resistance could be intensified. Without the empathetic understanding and support that caring managers provide, workers may face heightened feelings of alienation and frustration under the impersonal control of algorithmic systems (Wang et al., 2021). This can potentially escalate their resistance behaviors as they struggle to cope with the emotional distress caused by unfair treatment (Wiener et al., 2023). For example, a gig worker who feels unfairly compensated by an algorithmic system, yet lacks a supportive manager to address these concerns, might become increasingly resistant, perhaps by openly criticizing the platform or decreasing their work effort (Cotter, 2023). Thus, the negative association between algorithmic fairness and worker resistance could be less pronounced in situations where workers perceive a high level of caring from their human supervisors. Therefore, we propose the third hypothesis.

H3. Manager caring buffers the negative relationship between algorithmic fairness and worker resistance, such that this relationship is less pronounced when manager caring is greater.

3. Research method

Recognizing the limitations of cross-sectional data in establishing causal relationships and their susceptibility to common method variance (Maier et al., 2023), this research collected a longitudinal data to examine the proposed hypotheses in Fig. 1. We targeted the riders on food-delivery platforms because of the prevalence of algorithmic systems in managing these gig workers, making them a suitable subject to explore the impact of algorithmic management on worker behaviors (Cram et al., 2022). In particular, the food-delivery platforms in China (e.g., Meituan and Eleme) provide a unique context to examine the role of human managers in the algorithm-driven platform governance. These platforms are prevalent in urban areas and feature numerous delivery stations, each managed by a station master (Zheng & Wu, 2022). This arrangement creates a hybrid management mode where delivery riders are guided by both algorithmic directives and human oversight. The station masters, serving as human managers, offer essential support and solicitude, help in clarifying algorithmic decisions, and advocate for the workers' rights and interests (Wu et al., 2023). The co-existence of algorithmic control and human supervision in this context provides a suitable opportunity to explore the managerial synergy between humans and algorithms, examining how they intersect to shape worker resistance.

To obtain a representative sample of food-delivery riders for our study, we utilized Credamo, a premier online survey platform in China. Credamo is akin to well-known platforms such as MTurk and Qualtrics in its operation and is distinguished by its substantial user base, comprising over three million registered individuals. This extensive reach and its demographic diversity made Credamo a suitable choice for accessing a broad spectrum of food-delivery riders from numerous areas throughout China. The platform is especially noted for its high credibility and reliability, attributes that have led to its frequent utilization in scholarly research, as evidenced by its citations in top-tier IS journals (e.g., Hong et al., 2023; Qiu et al., 2023; Tang et al., 2023). To ensure a diverse and unbiased dataset, potential respondents were randomly selected from its extensive user base. They were then systematically invited via the platform's dedicated sample service to participate in our structured

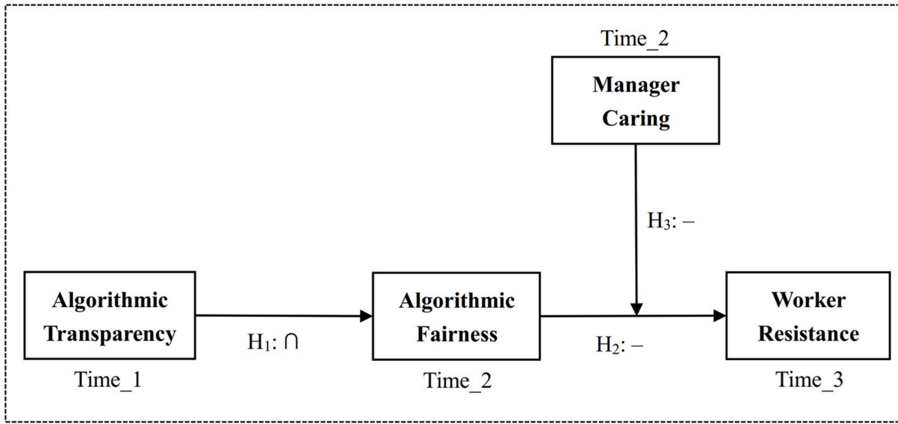

Fig. 1. Research model.

multi-stage survey.

Prior to conducting the formal survey, we carried out a pilot test. This phase involved a small sample of delivery riders ($n = 56$). The purpose of the pilot test was to evaluate the clarity, relevance, and potential misinterpretations of the survey items. Participants were also asked to provide feedback on any ambiguities or difficulties they encountered while answering the questions. Based on the insights gained from the pilot test, we made several adjustments to the questionnaire. These included rephrasing certain items that were reported to be overly abstract or susceptible to multiple interpretations, simplifying the language to enhance comprehensibility, and adjusting the scale of responses to better capture the intensity of respondents' perceptions and behaviors. We also added clarifying instructions at the beginning of the survey to ensure that respondents fully understood the context of our study.

The collection process was executed in three stages, with a two-week interval between each stage. In the first stage (Time_1), we collected data on algorithmic transparency from 1000 participants, and received 922 valid responses. Algorithmic transparency was assessed using a six-item scale adapted from Cram et al. (2022). In the second stage (Time_2), we collected data on algorithmic fairness and manager caring from the same 922 participants, and received 784 valid responses. Specifically, algorithmic fairness was measured using a four-item scale adapted from Wang et al. (2023), while manager caring was assessed using a five-item scale adapted from Micevski et al. (2017). In the final stage (Time_3), we collected data on worker resistance from the 784 respondents, and received 730 valid responses. Worker resistance was evaluated using a six-item scale drawn from Pregenzer et al. (2021a,b). All measurement items of the constructs were rated on seven-point Likert scales using "strongly disagree" and "strongly agree" anchors. An overview of the used items is shown in Table 1.

To ensure the identification consistency across stages, we collected the last four digits of the participants' mobile phone numbers for data matching. After matching, we obtained 676 food delivery riders as the final sample entering subsequent data analysis. To control for potential confounding effects, we also collected demographic information (i.e., gender, age, work tenure, average monthly income, and work type), as suggested by prior research (Cram et al., 2022; Wiener et al., 2023). In each stage, we also employed an Instructional Manipulation Checks (IMCs) item to detect problematic responses, which is "Please select strongly disagree for this item" (Paas et al., 2018). Collectively, invalid responses were detected by the following rules: 1) The completion time is outside the range of mean response time plus or minus 3 standard deviations; 2) respondents who failed to pass the IMCs; 3) respondents who reported inconsistent demographic information across stages (Meade & Craig, 2012). Table 2 presents the basic information of survey respondents in this study.

4. Results

The results of measurement model validation are provided in Table 3. The confirmatory factor analysis revealed a good fit to the data ($\chi^2/df = 2.87$, CFI = 0.93, TLI = 0.92, RMSEA = 0.05, SRMR = 0.04). All factor loadings were significant and above 0.60. The Composite Reliability (CR) values ranged from 0.84 to 0.91, exceeding the recommended threshold of 0.70. The Cronbach's alpha values for algorithmic transparency ($\alpha = 0.86$), algorithmic fairness ($\alpha = 0.81$), worker resistance ($\alpha = 0.83$), and manager caring ($\alpha = 0.88$) exceeded the recommended threshold of 0.70, indicating good internal consistency and reliability. The Average Variance Extracted (AVE) values ranged from 0.51 to 0.66, surpassing the 0.50 benchmark, confirming convergent validity. Moreover, Table 4 shows that the square root of the AVE for each construct was found to be greater than its correlations with other constructs, supporting the discriminant validity among the focused constructs. In summary, the results of measurement testing provided evidence for the reliability, convergent validity, discriminant validity, and overall fit of the measurement model. These results laid a solid foundation for the testing of the proposed hypotheses, as detailed in the following.

H1 proposes an inverted U-shaped relationship between algorithmic transparency (AT) and algorithmic fairness (AF). To establish an inverted U-shaped relationship, three conditions need to be satisfied (Haans et al., 2016): (1) the quadratic term must be significantly negative; (2) the slope must be sufficiently steep at both ends of the data range, meaning a sufficiently positive slope when AT is low (mean minus one SD) and a sufficiently negative slope when AT is high (mean plus one SD); (3) the turning point must be located well within the data range. Table 5 presents the results of quadratic regressions. The quadratic term (AT^2) was found to be significant and negative ($\beta = -0.187$, $p < 0.01$), satisfying the first criterion. The slope analysis showed a sufficiently positive slope when AT was low (slope $= 0.907$, $p < 0.001$) and a sufficiently negative slope when AT was high (slope $= -0.227$, $p < 0.05$), meeting the second criterion. The inflection point was calculated at a value of AT = 5.12, well within the data range (1–7), fulfilling the third criterion. Therefore, H1 was supported. Fig. 2 illustrates the inverted U-shaped pattern, showing that as AT increases, AF first rises to a peak and then starts to decline.

H2 proposes that algorithmic fairness (AF) is negatively associated with worker resistance (WR). Table 5 shows that AF can negatively predict WR ($\beta = -0.476$, $p < 0.001$), offering evidence for H2. Furthermore, H1 and H2 collectively implies that AT may have a U-shaped relationship with WR mediated by AF. The results indeed demonstrated a U-shaped relationship when we regressed WR on AT and its quadratic term, because 1) the coefficient of AT^2 is significantly positive ($\beta = 0.272$, $p < 0.01$); (2) a sufficiently negative slope when AT

Table 1
Measurement items used in this study.

Constructs	No.	Items
Algorithmic transparency (Cram et al., 2022)	AT1	The platform specifies exactly which types of data it collects from me to make decisions about my work.
	AT2	The platform provides detailed descriptions of how my data is used and protected in relation to the algorithm.
	AT3	The platform keeps me informed about updates to its algorithm and explains how these changes could affect my work.
	AT4	The platform uses clear examples or scenarios to show how its algorithm makes decisions affecting my tasks.
	AT5	The platform clearly communicates the main factors that its algorithm uses to determine outcomes that affect me.
	AT6	The platform gives specific advice on how I can improve my performance based on feedback from the algorithm.
Algorithmic fairness (Wang et al., 2023)	AF1	The platform's algorithm distributes desirable tasks evenly among workers, ensuring equal opportunity.
	AF2	The platform's algorithm allocates rewards and penalties fairly and equally across all workers.
	AF3	The platform's algorithm consistently applies the same rules to all workers when allocating tasks and evaluating performance.
	AF4	The platform includes mechanisms that allow worker feedback to influence decisions made by the algorithm.
Worker resistance (Pregenzer et al., 2021a, b)	WR1	I choose to accept or decline tasks assigned by the algorithm based on my schedule and preferences.
	WR2	I employ various strategies to avoid constant real-time monitoring by the algorithm.
	WR3	I sometimes provide incorrect information to the algorithm to protect my interests.
	WR4	I look for and use weaknesses in the algorithm to my advantage.
	WR5	I directly contact customers to discuss and possibly improve their reviews.
	WR6	I join online forums and groups to learn techniques for influencing the algorithm's decisions.
Manager caring (Micevski et al., 2017)	MC1	My station master shows genuine concern for my personal well-being, not just my work output.
	MC2	My station master actively listens to my views on work issues and tries to understand my perspective.
	MC3	My station master provides direct help to solve challenges I face at work.
	MC4	My station master ensures I have the resources and support needed to do my job well.
	MC5	My station master understands my feelings and work experiences, which makes me feel valued.

Note: The station master is the supervisor of delivery workers in the context of the food-delivery platform. WR1 was excluded from data analysis because it represents a reasonable action permitted by gig platforms.

is low (slope = −1.135, $p < 0.01$) and a sufficiently positive slope when AT is high (slope = 0.514, $p < 0.05$); and (3) the turning point (AT = 4.78) locates within the data range (1–7). Fig. 3 depicts the relationship pattern between AT and WR, indicating that AT can increase or decrease WR, depending on the level of AT relative to the turning point. The results of mediation analysis show that the bootstrapped 95% confidence interval of the indirect effect of algorithmic transparency on worker resistance does not include zero (indirect effect = 0.089, 95% CI = [0.051, 0.122]), indicating that the mediating effect of algorithmic fairness is significant (Zhao et al., 2010).

Table 2
Demographic information of survey participants (n = 676).

Variable	Category	Frequency	Percentage (%)
Gender	Male	629	93
	Female	47	7
Age	<30 years	304	45
	31–40 years	216	32
	41–50 years	135	20
	>50 years	21	3
Monthly income	<3000 RMB	54	8
	3000-5000 RMB	176	26
	5000-10000 RMB	372	55
	>10000 RMB	74	11
Work type	Part-time	149	22
	Full-time	527	78
Work tenure	<6 months	60	9
	6–12 months	122	18
	1–2 years	237	35
	2–4 years	162	24
	>4 years	95	14
Gig platform	Meituan	257	38
	Eleme	237	35
	Daojia	68	10
	Hema	80	12
	Others	34	5

Table 3
The results of the measurement model.

Constructs	Items	Loadings	CR	AVE
Algorithmic transparency	AT1	0.655	0.871	0.530
	AT2	0.743		
	AT3	0.765		
	AT4	0.681		
	AT5	0.790		
	AT6	0.727		
Algorithmic fairness	AF1	0.812	0.849	0.584
	AF2	0.748		
	AF3	0.769		
	AF4	0.725		
Worker resistance	WR2	0.720	0.838	0.509
	WR3	0.653		
	WR4	0.728		
	WR5	0.765		
	WR6	0.696		
Manager caring	MC1	0.821	0.907	0.660
	MC2	0.863		
	MC3	0.780		
	MC4	0.795		
	MC5	0.802		

Table 4
Descriptive statistics of the studied variables.

Variables	M	SD	1	2	3	4
1 Algorithmic transparency	4.211	1.516	**0.728**			
2 Algorithmic fairness	3.948	1.337	0.295	**0.764**		
3 Worker resistance	4.232	1.640	−0.310	−0.419	**0.713**	
4 Manager caring	3.825	1.019	0.094	0.182	−0.267	**0.812**

Note: bold-faced diagonal elements are the square roots of AVEs.

H3 proposes that manager caring can buffer the negative association between algorithmic fairness and worker resistance. The results revealed a significant interaction term between algorithmic fairness and manager caring ($\beta = 0.301$, $p < 0.05$). The moderating effect was further probed using simple slope analysis. We counted the level that is one standard deviation above the mean of manager caring as high managerial caring, and the level that is one standard deviation below the mean of manager caring as low managerial caring. Here, the mean of manager caring is zero because we centered this variable to reduce

Table 5
The results of hypothesis testing.

Variables	Worker resistance	Algorithmic fairness	Worker resistance
Constant	7.945** (2.631)	1.064** (0.407)	5.250* (2.184)
Gender	0.048 (0.056)	−0.102 (0.061)	−0.066 (0.075)
Age	−0.067 (0.055)	0.089 (0.050)	−0.161* (0.080)
Monthly income	0.152** (0.057)	0.158* (0.065)	−0.075 (0.079)
Work type	0.101 (0.080)	−0.134 (0.084)	0.189* (0.088)
Work tenure	−0.198* (0.096)	0.210* (0.098)	−0.343*** (0.096)
Algorithmic transparenct (AT)	−2.601** (0.863)	1.915* (0.782)	−2.005 (1.318)
AT^2	0.272** (0.089)	−0.187** (0.065)	0.124 (0.073)
Algorithmic fairness (AF)			−0.476*** (0.099)
Manager caring (MC)			−0.158 (0.087)
AF*MC			0.301* (0.144)
R-square	0.433	0.467	0.595

Note: Unstandardized regression coefficients and standard errors are reported; $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

multicollinearity issue in regression analysis (Dawson, 2014). Thus, the level representing high (low) manager caring is (minus) one standard deviation (SD = 1.019). Then we calculated the slope under the condition of high (low) manager caring by adding (subtracting) the main effect coefficient to the product of the interaction coefficient and the standard deviation. The results showed that the slope in the high condition is −0.169, and the slope in the low condition is −0.783, given the slope in the middle condition is −0.476. This trend indicates that, as the level of manager caring increases, the negative association between algorithmic fairness and worker resistance would gradually become weaker. Therefore, manager caring can buffer the negative relationship between them, supporting H3. Fig. 4 illustrates the buffering role of manager caring.

5. Discussion

This research explores the intricate interplay between algorithmic transparency, fairness, and resistance in the gig economy. It identified a curvilinear pattern: initial increases in algorithmic transparency enhance fairness perception and reduce resistance among gig workers, but beyond a certain threshold, further transparency paradoxically decreases fairness perception and increases resistance. Moreover, the findings highlight the buffering role of manager caring, indicating that empathetic and supportive actions from human supervisors can alleviate workers' negative responses to algorithmic decisions. These findings not only challenge the conventional belief that enhancing transparency is always beneficial but also provide new perspectives for the design and implementation of algorithmic systems on gig platforms, emphasizing the importance of balancing efficient algorithmic governance with empathetic human intervention. Next, we discuss the theoretical contributions as well as practical implications in detail.

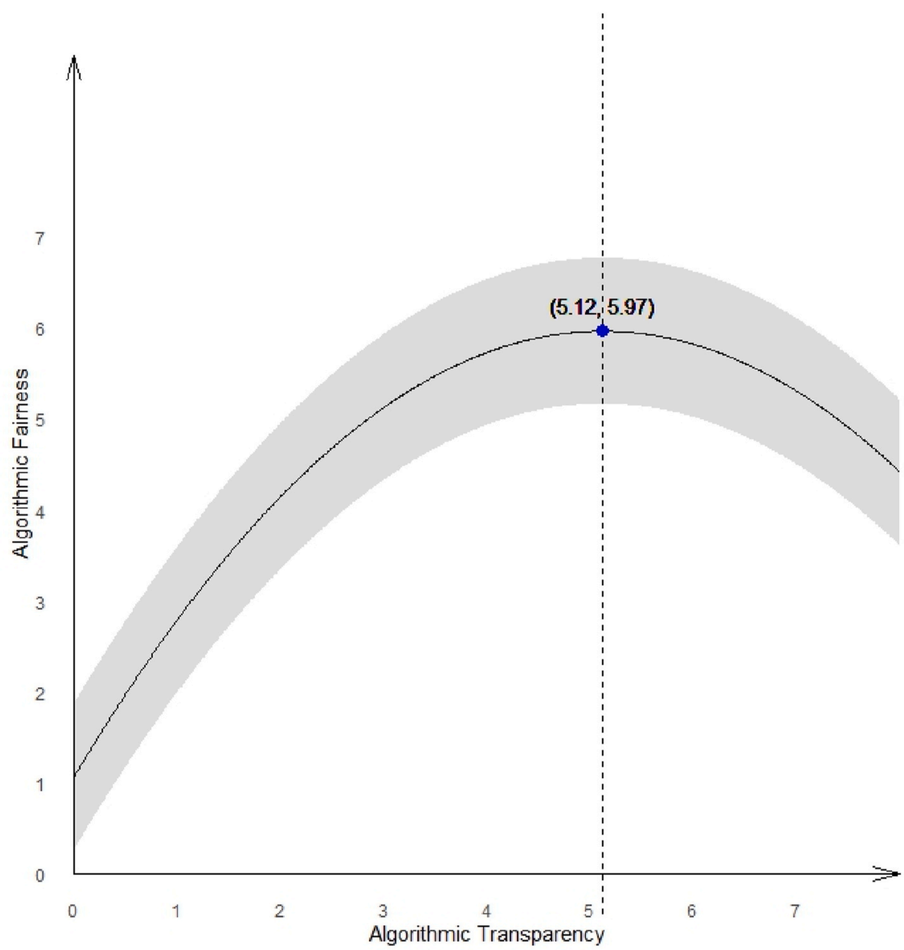

Fig. 2. The nonlinear relationship between algorithmic transparency and fairness.

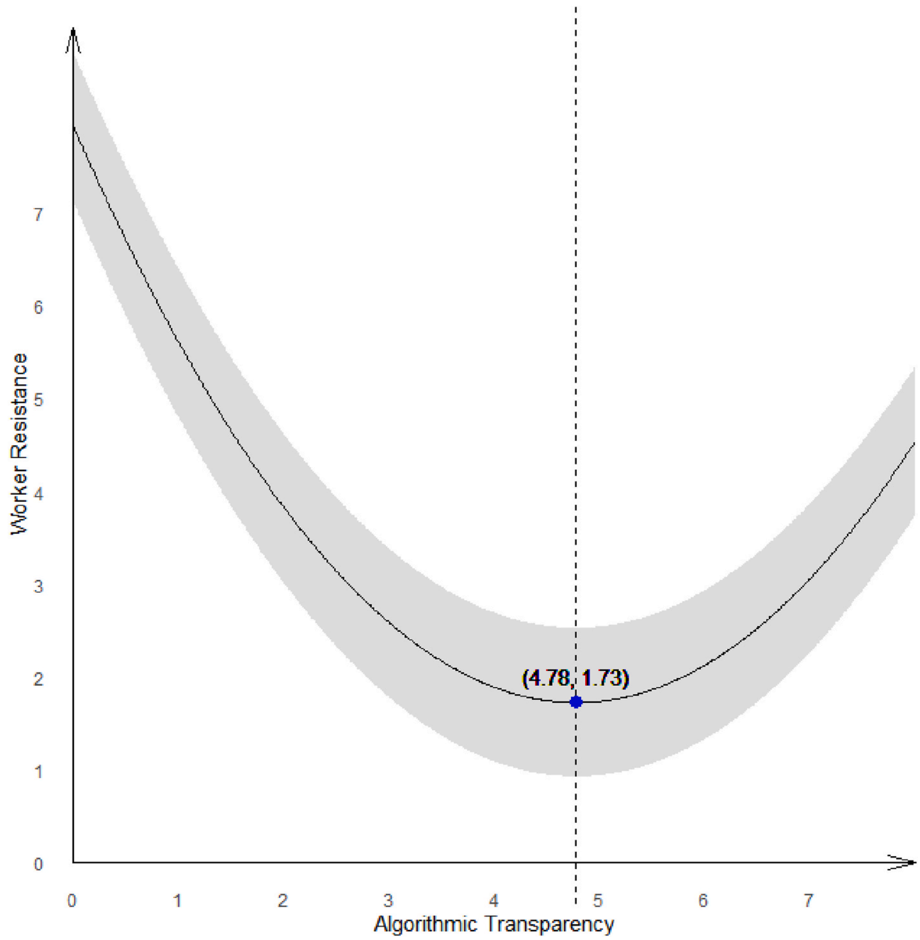

Fig. 3. The nonlinear relationship between algorithmic transparency and worker resistance.

5.1. Theoretical contributions

The primary contribution of this research is the empirical demonstration of the inverted U-shaped relationship between algorithmic transparency and worker resistance in the gig economy. This finding challenges the prevailing notion that greater transparency in algorithmic systems fosters positive attitudes and behaviors of those affected by the systems. Existing literature has emphasized the benefits of transparency, advocating for its role in enhanced trust, satisfaction, and acceptance in various contexts (Aysolmaz et al., 2023; Lee & Cha, 2023; Liu, 2021; Suen & Hung, 2023; Wanner et al., 2022). However, our research adds a new dimension to this discourse by revealing that beyond a certain threshold, heightened transparency can paradoxically lead to increased worker resistance. This suggests that while a certain degree of transparency is necessary to build trust and understanding, excessive transparency may overwhelm workers, potentially leading to confusion and information overload. Such speculation about the negative effects of excessive transparency, while informed by cognitive processing literature (Langer & König, 2023; Wang et al., 2023; You et al., 2022), requires further empirical investigation. This dark aspect of transparency pushes the boundaries of our understanding beyond simplistic linear models. It thus calls for a re-examination of transparency practices in algorithm design and highlights the need for a balanced approach in future implementations.

Second, our research empirically unveils the pivotal role of human intervention, specifically manager caring, in shaping worker responses to algorithmic systems. While prior research has focused largely on the operational capabilities of algorithmic systems (Möhlmann et al., 2021; Kinowska & Sienkiewicz, 2023; Won et al., 2023), it often overlooks the human aspect of managing workers under these systems. Our findings demonstrate that manager caring can mitigate the adverse effects of perceived algorithmic unfairness on worker resistance. This suggests that even in highly automated environments, the presence and actions of human managers are instrumental in shaping workers' perceptions and responses. This observation was directly supported by our data, adding a new layer to the discourse on algorithmic management. Furthermore, this finding introduces the concept of "Human-Algorithm Co-management" model, where the efficiency of algorithmic systems is complemented by the nuanced understanding and emotional intelligence of human managers. While the broader implications of this hybrid management model are informed by the literature and call for further empirical investigation (Benlian et al., 2022; Claure et al., 2023), they suggest a promising direction for bridging a critical gap in our understanding of algorithmic management and human-machine interaction.

Lastly, this research empirically elucidates the nuanced ways in which resistance behaviors are linked to the transparency of algorithmic systems from the lens of fairness perception. This advances the understanding of the role fairness played in the relationship between algorithmic features and worker reactions in the context of the gig economy. Specifically, our data demonstrate an inverted U-shaped curve where increased algorithmic transparency initially boosts perceived fairness but then leads to a decrease beyond a certain point. This decline in perceived fairness, potentially due to information overload, is informed by existing research (de Fine Licht & de Fine Licht, 2020) but remains speculative and requires further empirical validation. Furthermore, we establish that higher levels of perceived fairness are associated with lower levels of worker resistance. Workers are less likely to resist when they perceive the algorithms governing their work as fair. This finding

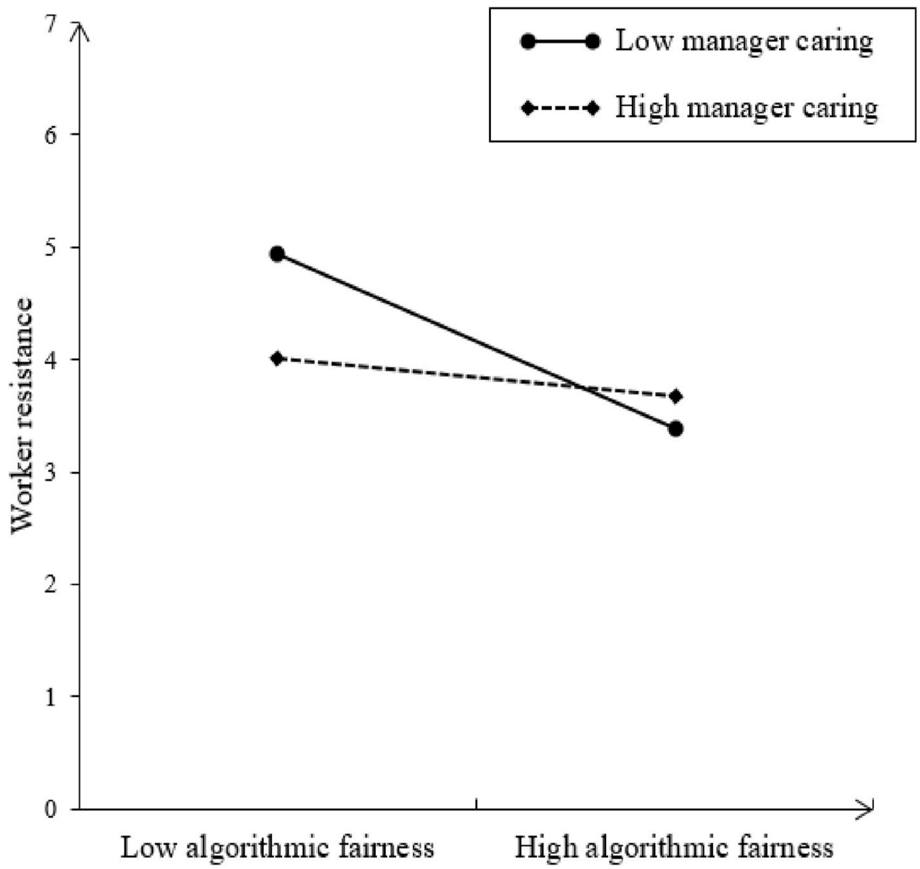

Fig. 4. The buffering role of manager caring.

offers a nuanced understanding of how fairness perception can drive workers' behavioral responses to algorithmic systems (Lavanchy et al., 2023; Starke et al., 2022). It highlights the importance of managing not just the transparency of algorithmic processes but also how these processes are perceived by workers (Wang et al., 2023). By illuminating the mediating role of perceived fairness, this research offers a deeper understanding of how algorithmic transparency can be effectively leveraged to foster a more positive and cooperative work environment.

5.2. Practical implications

First, our empirical data reveals a curvilinear relationship between algorithmic transparency and worker resistance, suggesting that food delivery platforms should provide enough information about their processes to foster trust and fairness among workers. However, the level of detail should be balanced to avoid information overload, which can inadvertently increase resistance. Notably, information overload is an interpretation informed by the literature and thus should be considered speculative until further validated. For practical implementation, we recommend these platforms clearly communicate the key factors influencing ride allocations and fare calculations in a concise manner. Implementing user-friendly interfaces that succinctly explain these processes may help demystify the algorithmic systems and enhance worker trust. While these strategies are designed to improve worker satisfaction and enhance operational efficiency, the direct impact of such measures should be empirically tested in future studies to verify their effectiveness across different algorithmic work contexts.

Second, building on our empirical finding that workers' perceptions of algorithmic fairness significantly decrease their resistance, we recommend gig platforms focus on enhancing the fairness in algorithmic systems. This involves not only ensuring equitable outcomes but also clearly explaining the decision-making processes. For example, a food delivery service could improve transparency by outlining how delivery assignments correlate with factors such as location proximity and order volume, helping delivery personnel understand the logic behind task distributions. Additionally, establishing robust feedback mechanisms where workers can express concerns or dispute decisions may enhance perceived fairness. Implementing a system where workers can flag perceived unfair instances for review by a human team not only bolsters fairness but also fosters a participatory environment, reducing feelings of alienation and resistance. By addressing fairness in outcomes and processes, gig platforms can cultivate a work climate of equity and justice, boosting worker motivation and strengthening the relationship between workers and platforms. These strategies, grounded in our empirical analyses, suggest effective ways to improve working conditions in the gig economy, enhancing operational efficiency and worker satisfaction.

Third, the data from our study elucidates the buffering role of manager caring between perceived fairness and worker resistance, emphasizing the importance of human elements in predominantly algorithm-driven environments. This finding suggests that investing in training managerial staff to better understand and empathize with gig workers' challenges can mitigate resistance. For example, training programs could be developed to enhance managers' communication skills and their ability to provide personalized support. Practical implementations might include regular check-ins with workers and guidance on navigating the algorithmic system. Additionally, initiatives like flexible working arrangements or mental health support services could illustrate a platform's concern for workers' well-being, such as a system alerting managers to drivers at risk of overworking to prompt supportive interventions. However, while our findings demonstrate the value of managerial care in reducing resistance within the food delivery context, generalizing these results to other algorithmic work contexts demands a nuanced approach. The specific role and impact of human managers can vary significantly across different sectors due to distinct operational and interactional dynamics inherent to each industry. For instance, in sectors like ride-hailing, where algorithmic efficiency and real-time decision-making are paramount, the introduction of human managerial elements might not provide the same benefits as observed in food delivery. Similarly, in highly specialized fields such as stock trading or medical diagnostics, where decisions are often based on complex data analysis and require high levels of precision and speed, the role of human managers might shift towards oversight and ethical governance rather than direct involvement in algorithmic decisions. Therefore, while managerial care has shown benefits in the food delivery sector, its applicability to other contexts where human interaction plays a less critical role in daily operations remains speculative. More empirical research is needed to delineate the conditions under which human interventions can effectively complement algorithmic decision-making across diverse sectors.

Finally, the implications of this research for policymaking in the gig economy are also significant, suggesting that policymakers should adopt regulations ensuring a reasonable degree of transparency in algorithmic decision-making to make it meaningful and manageable for workers. For example, policies could require platforms to disclose key elements of their algorithmic processes in a format easily understandable to non-technical audiences, thereby enhancing trust and fairness perceptions. Additionally, regulations should focus on safeguarding fairness within these systems, potentially through the establishment of independent bodies tasked with overseeing and auditing algorithmic practices for fairness and transparency, adding an essential layer of accountability. These policy suggestions, informed by our empirical findings, propose a collaborative approach where platforms, workers, and regulators work together to create a gig economy that benefits all stakeholders, addressing current challenges while anticipating future needs to maintain the viability and fairness of gig employment.

5.3. Limitations and future research

First, this study has not included each variable of interest at all three measurement points. This prevented us from applying complex analytical methods, such as the Random Intercept Cross-Lagged Panel Model (RI-CLPM) (Mulder & Hamaker, 2021), which would more substantially enable causal inference about the complex relationships between algorithmic transparency, fairness, and resistance over time. Future research can capture all relevant variables across multiple time points to enable the use of RI-CLPM or similar techniques, thus providing more definitive insights into how changes in algorithmic transparency impact worker behavior and well-being over time. Such efforts would enhance our understanding of the dynamic interplay between technology and human behavior in the evolving landscape of the gig economy.

Second, this study measured algorithmic transparency using subjective assessments from gig workers, which may introduce individual differences in how transparency is perceived. While this approach provides insights into workers' personal experiences, it might not fully capture the objective transparency of algorithms used across different platforms. Future research could strengthen our findings by incorporating objective measures of algorithmic transparency. Such measures could include the accessibility and clarity of information provided by platforms about their algorithmic processes. This could be assessed through independent audits or by employing standardized transparency metrics developed in the field. By integrating subjective and objective assessments, researchers can more accurately discern the impact of actual versus perceived transparency on worker behavior and well-being. These efforts would advance our understanding of the nuanced effects of transparency in algorithm-driven work environments.

Finally, transparency is conceptualized primarily as the transparency of process in this study. However, transparency can encompass multiple dimensions, such as transparency of outcomes and the transparency of agents' ability to affect the system. Each of these dimensions could differently impact workers' perceptions of fairness and their subsequent resistance, which were not distinctly measured in our study. Similarly, worker resistance may also manifest in varied forms, and this study primarily captured this in aggregate rather than in its nuanced expressions. We admit that this lack of granularity could limit the specificity of our findings, as different forms of resistance may be influenced differently by various aspects of transparency. Future research can dissect these constructs more finely and explore the specific dimensions of transparency and resistance. Such research could significantly enhance our understanding of these dynamics and inform more tailored managerial strategies and algorithmic designs.

6. Conclusion

This research focuses on how workers perceive and react to the transparency of algorithmic systems in the gig economy, revealing a curvilinear relationship where increased transparency initially enhances fairness and reduces resistance. However, beyond a certain point, further transparency paradoxically diminishes fairness and heightens resistance. This challenges the traditional view that more transparency always yields positive outcomes, highlighting the necessity for a balanced approach. Additionally, we found that empathetic actions by managers are crucial in mitigating resistance, emphasizing the potential of integrating human elements like managerial caring into algorithm-driven environments. These insights collectively support a human-algorithm co-management mode wherein the advantage of AI efficiency and human empathy are both leveraged. This emphasizes creating work environments where algorithmic systems and human elements are seamlessly integrated, ensuring both technological effectiveness and human well-being. In conclusion, this research marks a significant step towards improving the interface between technology and human behavior, a cornerstone of the ''Ideal Machine-Human Experience'' in the algorithmization of work.

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CRediT authorship contribution statement

Peng Hu: Writing – original draft, Investigation, Formal analysis, Conceptualization. **Yu Zeng:** Writing – review & editing, Resources, Methodology, Data curation. **Dong Wang:** Writing – review & editing, Validation, Project administration. **Han Teng:** Methodology, Resources, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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